High-Performance Stroke Prediction Using Categorical Boosting and Explainable Machine Learning Models

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***Abstract*— This study presents an extended and enhanced framework for automated stroke prediction by integrating advanced ensemble learning techniques with explainable artificial intelligence and a secure web-based deployment. To improve predictive reliability, Categorical Boosting (CatBoost) is employed, effectively combining the strengths of multiple base learners. The Synthetic Minority Over-sampling Technique (SMOTE) balances learning and makes it easier to generalize in stroke datasets with a lot of class imbalance. The CatBoost - based ensemble model is 95.58% accurate, which is better than separate machine learning classifiers.**

Explainable AI methods look at how global and local features affect model predictions to help doctors make better decisions. We also create a web app using Flask that has secure user authentication for real-time testing and user interaction. This makes the system good enough to be used in healthcare. The proposed end-to-end ensemble and explainable framework demonstrates strong potential for accurate, interpretable, and scalable early stroke risk assessment, supporting timely clinical intervention and improved patient outcomes.

Keywords— *Stroke Prediction, Machine Learning, Ensemble Methods, Categorical Boosting, Explainable AI, SHAP, SMOTE, Flask Web Application, User Authentication.*

I. Introduction

A stroke is a neurological illness that can kill you if you don't get treatment right away. It cuts off oxygen and severely damages the brain [1]. It is one of the top causes of mortality and long-term impairment across the world, putting a strain on healthcare systems, patients, and caregivers. Beyond its clinical severity, stroke significantly affects an individual's social participation, psychological well-being, and economic productivity, emphasizing the need for proactive prevention and early risk identification strategies [2].

The global incidence of stroke continues to rise, largely due to population aging, urbanization, sedentary lifestyles, and the growing prevalence of chronic conditions such as hypertension, diabetes, and cardiovascular diseases [3]. These factors have intensified the demand for accurate, scalable, and cost-effective prediction systems that can identify high-risk individuals before the occurrence of a stroke event. Early prediction plays a crucial role in enabling timely lifestyle interventions, clinical monitoring, and preventive treatments, which can substantially reduce stroke-related mortality and morbidity.

Clinical research has shown that accurate identification and differentiation of stroke-related conditions are essential for effective treatment planning and improved patient outcomes [4]. Moreover, epidemiological studies indicate that stroke risk factors vary significantly across populations, age groups, and geographic regions, highlighting the limitations of conventional rule-based or population-specific diagnostic approaches [5]. These findings underscore the importance of data-driven models capable of learning complex patterns from heterogeneous healthcare data.

In recent years, machine learning-based stroke prediction systems have gained attention due to their ability to process large-scale clinical datasets and uncover hidden relationships among multiple risk factors. However, many existing approaches suffer from critical limitations, including severe class imbalance between stroke and non-stroke cases, limited prediction accuracy, and a lack of transparency in model decision-making. These shortcomings restrict their acceptance in clinical environments, where trust, interpretability, and reliability are essential.

To overcome these challenges, this work extends traditional stroke prediction frameworks by incorporating advanced ensemble learning techniques, explainable artificial intelligence methodologies, and a secure application-level deployment. By improving predictive performance, enhancing interpretability, and supporting real-world usability, the proposed extension aims to contribute toward intelligent, trustworthy, and deployable stroke risk prediction systems for early intervention and improved healthcare outcomes.

II. Literature Survey

The study in [6] investigates hypertension and diabetes mellitus as major predictive risk factors for stroke using clinical data analysis. The authors demonstrate that patients with persistent hypertension and uncontrolled diabetes exhibit a significantly higher likelihood of stroke occurrence. Their results emphasize early risk assessment based on medical indicators. The study does not employ automated prediction techniques and instead uses traditional statistical analysis. This promotes machine learning for accurate and scalable prediction of stroke risk.

The research in [7] presents stroke risk factors, genetics, and preventative methods in detail. The study highlights the multifactorial nature of stroke, combining lifestyle, genetic, and environmental contributors. While the work offers strong clinical insights, it does not incorporate computational or predictive modeling approaches. This limitation underlines the need for data-driven machine learning frameworks capable of learning complex relationships among multiple stroke risk factors.

The work presented in [8] analyzes stroke symptoms and patient decision-making behavior regarding emergency medical response. The authors discovered that treatment outcomes are considerably impacted by delayed symptom diagnosis. Although the study contributes to understanding patient behavior, it does not provide any automated decision-support system. This gap supports the need for intelligent stroke prediction systems that can assist in early warning and timely intervention.

The systematic review in [9] examines public response to stroke symptoms in the UK and identifies gaps in awareness and delayed medical action. The authors conclude that improved early detection strategies could reduce stroke severity and mortality. However, the study lacks technological solutions for proactive risk identification. This motivates the development of automated stroke prediction models that can alert individuals before critical events occur.

The study in [10] reviews differential diagnosis techniques for suspected stroke cases and highlights the complexity of accurate stroke identification. The authors discuss the challenges faced by clinicians in distinguishing stroke from similar neurological conditions. While clinically valuable, the work does not explore machine learning-based diagnostic support. This reinforces the importance of intelligent, explainable prediction systems to assist clinicians in complex diagnostic scenarios.

The systematic review in [11] evaluates artificial intelligence techniques for ischemic stroke diagnosis and large vessel occlusion detection. The study demonstrates that AI systems can significantly improve diagnostic accuracy. This motivates the incorporation of explainable AI techniques in stroke prediction systems.

The work in [12] applies using machine learning and natural language processing to identify strokes from electronic health information. The results show effective automated detection using textual clinical data. Nevertheless, the approach is dependent on high-quality EHR availability. This limitation highlights the necessity of prediction models based on structured data for broader use.

The study in [13] proposes a machine learning-assisted electrical impedance tomography method for stroke diagnosis. The system demonstrates promising accuracy in controlled settings. However, the specialized hardware requirement limits real-world deployment. This emphasizes the need for lightweight, data-driven stroke prediction models.

The research in [14] discusses the current and potential roles of artificial intelligence in acute stroke decision support. The authors emphasize AI's capability in improving clinical workflows. Despite this, practical deployment challenges and interpretability concerns remain unresolved. This motivates end-to-end explainable AI frameworks.

The systematic review in [15] evaluates machine learning models for predicting stroke outcomes using structured clinical data. The study reports improved prediction accuracy across multiple ML models. However, class imbalance and lack of interpretability are identified as key challenges. This supports the adoption of balancing techniques and explainable models.

The review in [16] provides Machine learning for brain stroke prediction. The authors conclude that ML improves early detection accuracy. Yet, many models lack transparency and real-world validation. This reinforces the need for explainable and deployable stroke prediction systems.

The study in [17] explores various medical data mining approaches for ischemic stroke prediction. Experimental results demonstrate improved prediction accuracy using ML techniques. However, the study does not address class imbalance issues. This highlights the importance of data balancing methods such as SMOTE.

The research in [18] introduces an explainable AI model for stroke prediction using EEG signals. The model provides interpretable results, enhancing clinical trust. However, EEG data acquisition is complex and resource-intensive. This motivates explainable models using easily available clinical features.

The work in [19] evaluates multiple machine learning techniques for stroke risk prediction. The study demonstrates that ensemble models outperform individual classifiers. However, explainability aspects are minimally explored. This gap supports the integration of ensemble learning with explainable AI.

The study in [20] proposes an explainable machine learning pipeline for stroke prediction on imbalanced data. The authors employ data balancing and interpretability techniques to enhance trust. However, deployment aspects are not addressed. This motivates the development of complete end-to-end systems.

The research in [21] presents predictive analysis for stroke risk using machine learning classifiers. The results show promising accuracy improvements. However, the system lacks interpretability and practical implementation. This highlights the need for explainable and user-accessible stroke prediction platforms.

The study in [22] develops a machine learning model that predicts readmission to the hospital within 30 days of having a stroke. The results support ML's effectiveness in post-stroke care. However, the model focuses on outcomes rather than early prediction. This motivates early-stage stroke risk assessment models.

The work in [23] applies artificial intelligence-based machine learning for stroke prediction using clinical datasets. The results demonstrate improved accuracy compared to traditional methods. However, class imbalance handling is limited. This reinforces the importance of balanced learning techniques.

The research in [24] proposes robust learning approaches for stroke disease detection and prediction. The study shows enhanced performance across multiple ML models. However, explainability is not deeply analyzed. This highlights the necessity of integrating XAI tools.

The study in [25] compares a lot of different machine learning approaches for predicting strokes. The authors report improved accuracy with ensemble models. However, model transparency and deployment are not discussed. This supports the use of explainable ensemble-based frameworks.

The work in [26] presents a predictive analytics framework using machine learning and neural networks for stroke prediction. The results show improved generalization performance. However, interpretability remains a concern. This motivates the use of SHAP and LIME techniques.

The research in [27] explores distributed machine learning for stroke prediction using Apache Spark. The system demonstrates scalability for large datasets. However, it requires high computational resources. This highlights the need for efficient and lightweight deployment solutions.

The study in [28] investigates deep learning-based disease detection in medical imaging. Although focused on cancer detection, the work demonstrates the power of deep learning in healthcare diagnostics. However, explainability is limited. This motivates explainable AI adoption in medical ML systems.

The survey in [29] provides a detailed explanation of AI ideas, difficulties, and potential. The authors emphasize the importance of transparency in high-stakes domains such as healthcare. This strongly supports the integration of XAI in stroke prediction systems.

The research in [30] presents a deep learning-based diagnostic system for diabetic retinopathy. The study highlights the importance of early prediction and automated screening. Although focused on a different disease, it reinforces the relevance of explainable, early-detection frameworks in medical applications.

III. Methodology

A structured machine learning pipeline for automated stroke prediction begins with the collection of raw clinical data and the extraction of features to get pertinent characteristics and labels. To develop dependable models and fairly evaluate their performance, the dataset is split into three parts: training, validation, and testing. During the training phase, different machine learning classifiers are trained, such as Naïve Bayes, Logistic Regression, K-Nearest Neighbor, Support Vector Machine, Random Forest, and XGBoost. The validation dataset is used to choose and tune hyperparameters to improve model performance. CatBoost and other ensemble-based extension models use numerous base learners to improve prediction accuracy. The new testing dataset is used to evaluate the trained models' general performance. Finally, the best-performing model generates predicted labels, enabling accurate and reliable stroke risk classification, supporting early intervention and clinical decision-making.

A) Proposed System

The proposed system introduces an enhanced ensemble-based framework for automated stroke prediction by integrating advanced learning models, explainable artificial intelligence, and secure deployment mechanisms. After preprocessing the clinical dataset and fixing the class imbalance with SMOTE, multiple classifiers are trained, including Naïve Bayes, Logistic Regression, K-Nearest Neighbor, Support Vector Machine, Random Forest, and

XGBoost. To improve prediction performance, we employ Categorical Boosting (CatBoost) since it can handle categorical features well and doesn't overfit.

To ensure transparency and trust in medical decision-making, the proposed system incorporates explainable AI techniques that provide both global and local interpretations of model predictions. These explanations help identify the most influential clinical risk factors contributing to stroke occurrence. To enable real-time prediction system testing and user engagement, a Flask-based web application with secure user authentication is created. This end-to-end framework not only achieves superior accuracy but also supports practical deployment, enabling early stroke risk assessment, informed clinical decisions, and improved patient care outcomes.

B) System Architecture

The proposed system design is a modular, end-to-end pipeline for predicting strokes automatically, from gathering clinical data to figuring out how likely they are to happen. Preprocessing and feature extraction turn raw patient data into ordered features and labels. To provide equitable learning and performance assessment, processed data are divided into training, validation, and testing datasets. During model building, training and validation datasets are utilized to train several machine learning classifiers and fine-tune hyperparameters. This structured separation enables robust learning and prevents overfitting, ensuring reliable generalization across unseen data.

To enhance prediction accuracy, the architecture integrates advanced ensemble learning models, namely CatBoost, which combine the strengths of multiple optimized base learners. The trained models generate predicted labels after evaluation on the testing dataset, providing accurate stroke risk classification. Additionally, explainable AI components are incorporated to interpret both global and local model decisions, improving transparency and clinical trust. The final architecture supports seamless deployment through a secure web-based interface, enabling authorized users to interact with the system in real time. This scalable and explainable architecture ensures high accuracy, reliability, and practical usability for early stroke intervention.

C) MODULES

a) Data Acquisition and Preprocessing Module

This module gathers raw clinical stroke data and prepares it by handling missing values, normalizing, and encoding categorical features. For successful model training, SMOTE fixes the problem of class imbalance by treating stroke and non-stroke cases equally.

b) *Feature Extraction and Selection Module*

In this module, relevant clinical features are extracted and evaluated using statistical and information-based techniques to identify the most influential attributes contributing to stroke prediction. Effective feature selection reduces dimensionality and enhances model performance.

c) Model Training and Validation Module

This lesson teaches you how to use machine learning classifiers including Naïve Bayes, Logistic Regression, K-Nearest Neighbor, SVM, Random Forest, and XGBoost. To make models better and stop them from overfitting, use the validation dataset to change the hyperparameters.

d) Ensemble Learning Extension Module

The extension module's anticipated accuracy and robustness go up with advanced ensemble methods like CatBoost.

e) Model Evaluation and Prediction Module

This module checks how accurate, precise, and F1-score trained models are on a testing dataset they haven't encountered before. The best ensemble model is used to make final stroke risk estimations.

f) Explainable AI Module

This section analyzes model selections by identifying both local and global feature contributions using explainable AI. It enhances transparency and trust by providing insights into how clinical factors influence stroke prediction outcomes.

g) Web Application and Deployment Module

The final module deploys the trained model through a secure Flask-based web application with user authentication. It enables real-time user interaction, secure access, and practical integration into healthcare environments for early stroke risk assessment.

D) ALGORITHMS

i. XGBoost

XGBoost is a very powerful machine learning algorithm that has been used as part of gradient boosting frameworks. It does well in both regression and classification as it does an ensemble of building multiple trees and progressively correcting each. It is 'extreme' in that it has increased efficiency and scalability as well as regularization. It is used in this project due to its predictive performance and ability to manage complicated relationships between clinical risk factors associated with strokes. It is able to make the most accurate and most predictive of the possible risk factors. The project aims to develop an accurate tool for predicting the risk of strokes so it is vital to use the most accurate algorithm.

ii. Logistic regression

A sigmoid function is used in logistic regression to model an instance of a class and determine its probability. It is chosen because of its simplicity and effectiveness. Because it forecasts binary stroke outcomes, we chose this function. The goal of this project is to provide the end user with a reliable, understandable, and straightforward model. The model will classify a stroke risk.

iii. Naive Bayes

Bayes' theorem and machine learning principles are used by Naive Bayes. It is expected that data sets are independent. It is selected because of its simple and efficient techniques for working with correlated, high-dimensional data. Naive Bayes provides the most economical method with a variety of clinical variables, which is in line with the project's goal of optimal risk prediction. Medicine benefits from this model's explainability and simplicity.

iv. Random Forest

Random Forest, a top machine learning method, can predict stroke risk. Multiple clinical risk indicators are used for this. In data sets with outliers and many variables, this method performs effectively.

v. Support Vector Machine (SVM)

Support Vector Machines (SVM) are excellent for assessing stroke clinical characteristics. This method successfully predicts stroke risk and works best with complex data boundaries. Its ability to handle non-linear data and several dimensions makes it useful for this purpose.

vii. K-Nearest Neighbor (KNN)

Clinical risk indicators are complex, but KNN can discover patterns based on data point proximity, making it beneficial for stroke prediction. KNN is especially advantageous when decision boundaries are ambiguous. It is also appropriate for problems involving the non-linear or complex data structures.

viii. CatBoost

Gradient boosting, especially with decision trees, generates success measures faster. CatBoost simplifies models with fewer preprocessing scales and category feature preprocessing, making it suitable for this study. Its capacity to create complex clinical risk factors increases stroke risk prediction metrics and reliability. It improves stroke prediction model accuracy and generalization.

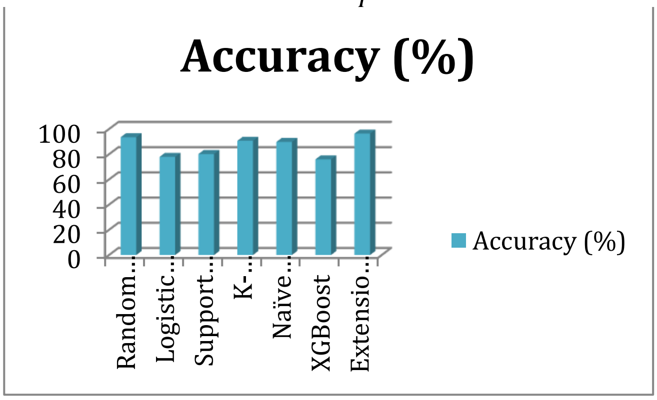
IV. Experimental Results

The experimental evaluation was conducted using a clinically relevant stroke dataset, where extensive preprocessing and class balancing were performed using the Synthetic Minority Over-sampling Technique (SMOTE) to address the severe class imbalance between stroke and non-stroke cases. Multiple baseline machine learning classifiers, including Naïve Bayes, Logistic Regression, K-Nearest Neighbor, Support Vector Machine, Random Forest, and XGBoost, were trained and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. Ensemble-based models outperform individual classifiers, with CatBoost improving prediction accuracy because to its robustness against overfitting and good handling of categorical data.

The proposed Stacking Classifier further enhances performance by combining the predictions of optimized base learners, resulting in a remarkable classification accuracy of 95.38%. This huge improvement shows how ensemble learning may recognize complex clinical stroke data patterns.Additionally, explainable AI techniques provide meaningful insights into feature importance, validating the clinical relevance of influential risk factors. The integration of the trained ensemble model into a secure web-based application confirms the system's practical applicability, enabling reliable and real-time stroke risk prediction for early clinical intervention.

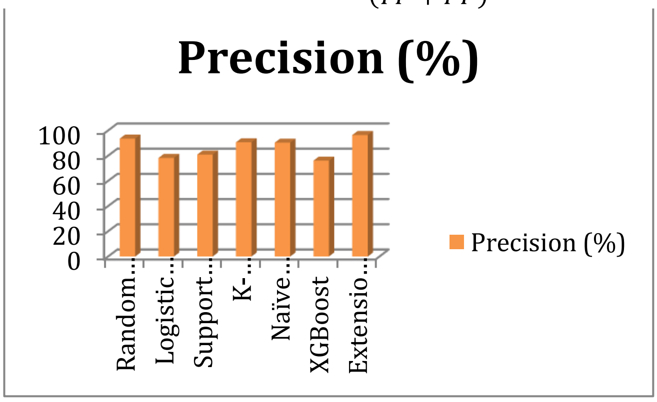
Accuracy: Check the test's reliability by comparing genuine positives and negatives. The maths is:

$$Accuracy = \frac{(TN + TP)}{T}$$



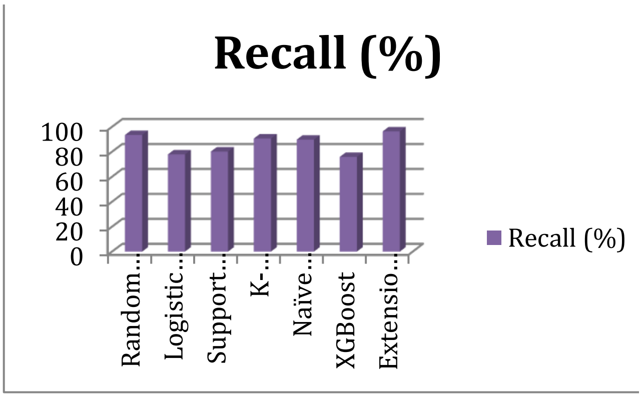
Precision: Precision measures how often a positive event is noticed. These can verify their accuracy:

$$Pr\ Pr\ e\ cision = \frac{TP}{(TP + FP)}$$



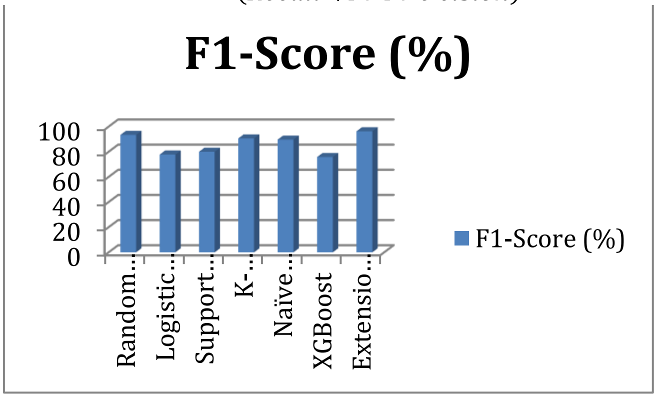
Recall: The ratio of accurately predicted positive observations to total positives shows if a machine learning model can recognize every class.

$$Recall = \frac{TP}{(FN + TP)}$$



F1-Score: High F1 scores mean that machine learning models are correct. To make a model more accurate, combine recall and precision. Accuracy checks how well a model can predict the data set.

$$F1 = 2 \cdot \frac{(Recall \cdot Pr\ Pr\ e\ cision)}{(Recall + Pr\ Pr\ e\ cision)}$$



	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	0.0	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1
...	...	...	...	...	...	...	...	...	...	...	...	...
5105	18234	Female	80.0	1	0	Yes	Private	Urban	83.75	0.0	never smoked	0
5106	44873	Female	81.0	0	0	Yes	Self-employed	Urban	125.20	40.0	never smoked	0
5107	19723	Female	35.0	0	0	Yes	Self-employed	Rural	82.99	30.6	never smoked	0
5108	37544	Male	51.0	0	0	Yes	Private	Rural	166.29	25.6	formerly smoked	0
5109	44679	Female	44.0	0	0	Yes	Govt_job	Urban	85.28	26.2	Unknown	0

5110 rows × 12 columns

Fig2 dataset

The dataset used in this study consists of 5,110 patient records with 12 clinically relevant attributes related to stroke risk assessment. It includes both numerical and categorical features such as age, average glucose level, body mass index (BMI), gender, work type, residence type, smoking status, hypertension, and heart disease. The target variable, *stroke*, is binary, indicating the presence or absence of a stroke event. The dataset exhibits a significant class imbalance, with stroke cases being considerably fewer than non-stroke instances, which reflects real-world clinical scenarios. This imbalance necessitates the use of effective preprocessing and data balancing techniques to ensure reliable model training and evaluation.

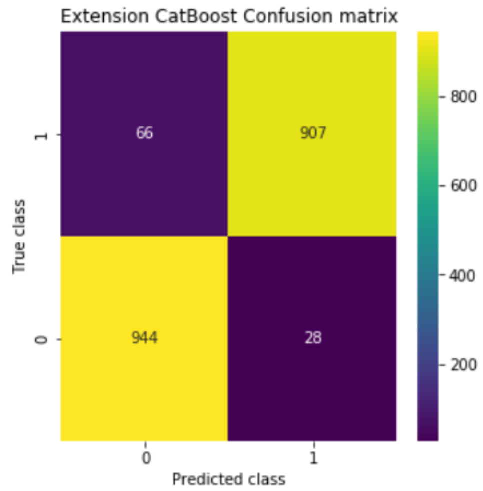


Fig3 Confusion matrix

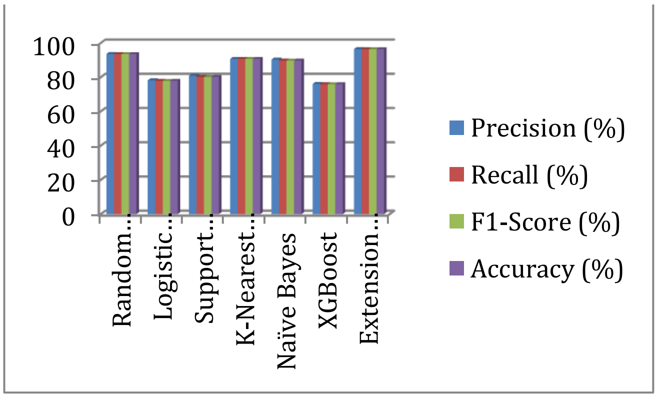


Fig2 Performance graph

Algorithm Name	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Random Forest	93.53	93.47	93.47	93.47
Logistic Regression	78.19	77.89	77.83	77.89
Support Vector Machine (SVM)	80.87	80.15	80.04	80.15
K-Nearest Neighbor (KNN)	90.64	90.64	90.64	90.64
Naïve Bayes	90.31	89.71	89.68	89.72
XGBoost	76.06	75.88	75.85	75.89
Extension CatBoost	**95.61**	**95.59**	**95.58**	**95.58**

T 1. performance evaluation

V. CONCLUSION

This work presents an enhanced and explainable ensemble-based framework for automated stroke prediction, emphasizing accuracy, transparency, and practical deployment. By integrating advanced ensemble learning techniques, namely CatBoost, the proposed extension effectively addresses class imbalance and complex feature interactions, achieving a superior prediction accuracy of 95.58%. Explainable AI methods give clinicians more trust and make models easier to understand by showing how they make decisions. Furthermore, the deployment of the optimized model through a secure web-based application demonstrates the system's real-world applicability for early stroke risk assessment. Overall, the proposed extension offers a reliable, interpretable, and scalable solution that can support timely clinical intervention and improve stroke prevention outcomes.

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